

Machine_Learning_Engineer — Step 3

Learn Machine Learning Algorithms

Detailed Learning Notes • CareerCompass

1. Learning Objective

Understand the major supervised and unsupervised algorithms, their assumptions, training process, strengths, limitations and appropriate use cases.

2. Core Concepts — Detailed Breakdown

Learning problem types

Separate supervised learning into regression and classification, and unsupervised learning into tasks such as clustering and dimensionality reduction. Define the target, features and business objective before selecting an algorithm.

Linear and logistic regression

Understand the model equation, loss functions, coefficients, regularization and interpretation. Logistic regression predicts class probabilities rather than directly outputting arbitrary class labels.

Tree-based methods

Learn decision trees, random forests and gradient boosting conceptually. Trees split data into regions; ensembles combine multiple models to improve generalization or predictive strength.

Distance and margin methods

Understand k-nearest neighbors and the intuition of support-vector machines. Feature scaling can be important because distances and margins depend on feature magnitudes.

Unsupervised learning

Study k-means clustering, hierarchical clustering and PCA. Understand that unsupervised results require interpretation because there may be no ground-truth labels.

Model assumptions and trade-offs

Compare interpretability, compute cost, data requirements, nonlinearity, robustness and calibration. There is no universally best algorithm.

3. Recommended Learning Workflow

- Define the prediction task and evaluation metric.
- Create a baseline model before complex models.
- Split data correctly and build preprocessing into the training pipeline.
- Compare several algorithms using the same evaluation protocol.
- Inspect errors and explain why the chosen model is appropriate.

5. Hands-on Project / Practice

Build a model comparison notebook for a classification dataset. Train logistic regression, a tree-based model and k-nearest neighbors; use a pipeline for preprocessing; compare precision, recall, F1 and ROC-AUC where appropriate; inspect misclassified examples; and document the trade-off between performance and interpretability.

6. Common Mistakes & Pitfalls

- Choosing an algorithm before defining the target and metric.
- Training preprocessing on the entire dataset before splitting.
- Comparing models on different test sets.
- Using accuracy for highly imbalanced classes without checking other metrics.
- Assuming a more complex model is automatically better.

7. Completion Checklist

- I can distinguish regression, classification and clustering.
- I can explain at least five common algorithms.
- I can choose metrics based on the problem.
- I can describe overfitting and regularization.
- I can compare models using a reproducible evaluation procedure.

8. Interview / Assessment Questions

- When would you prefer logistic regression to a tree ensemble?
- Why can k-NN be sensitive to feature scaling?
- What is regularization?
- Why is a baseline model important?
- What is the difference between training and inference?

9. Recommended References

Scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

Machine Learning in 2024 — freeCodeCamp: <https://www.youtube.com/watch?v=bmmQA8A-yUA>

Kaggle Learn: <https://www.kaggle.com/learn>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.